# Algebra & Functions — Timed Simulation 4

IBEW / ETA math section format

| 33 | 46 | ~84 sec | No |
|---|---|---|---|
| questions | minutes | per question | calculator |

## About this test

- All 33 questions from your photos of Timed Simulation 4, retyped clean and printable, in the original order. Wording, numbers and answer choices are kept as they appeared on screen.
- Graphs are redrawn on a clean grid to match what was on screen.
- Multiple choice, A–D. No calculator. No penalty for guessing — answer everything.

## How to take it

- Set a timer for 46 minutes and don't stop it.
- Skip anything that takes more than about 90 seconds and come back to it.
- The answer key with a short explanation for every question is at the back.

## What's on it

| | |
|---|---|
| Simplifying & factoring | Questions 4, 13, 22, 23, 24 |
| Number series | Questions 6, 7, 10, 18, 19 |
| Plugging in & solving | Questions 1, 15, 17, 20, 21, 28, 32 |
| Isolating a variable | Questions 26, 31 |
| Variable relationships (which statement is true) | Questions 8, 12, 14, 27, 30 |
| Same relationship: tables, graphs, words | Questions 2, 5, 25, 29 |
| Graphing & systems | Questions 3, 9, 11, 16, 33 |

# Question 1

Consider the following formula:

$2(B + A) - C = 5(C - B)$

**If $B = 4$ and $C = -3$, what is the value of $A$?**

A) 23

B) $-20$

C) $-8$

D) $-23$

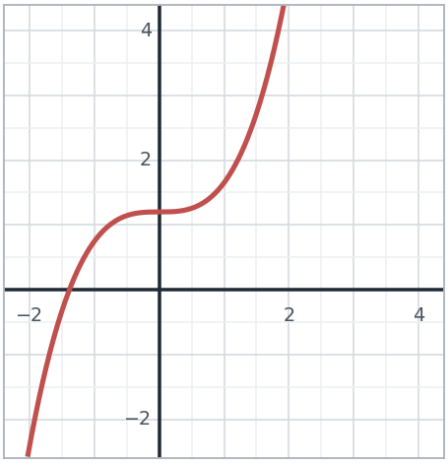
**WORKSPACE**

# Question 2

Consider the following table:

| x | y |
|---|---|
| 0.5 | 9.5 |
| 1 | 11 |
| 3 | 17 |
| 8 | 32 |

**Which of the following choices represents the same relationship as demonstrated in this table?**

A) 


B) Y is equal to the sum of ten times the value of x and a constant C, where C is equal to 1.

C)

| x | y |
|---|---|
| −5 | −7 |
| −3 | −1 |
| 0 | 8 |
| 5 | 23 |

D) None of the above

**WORKSPACE**

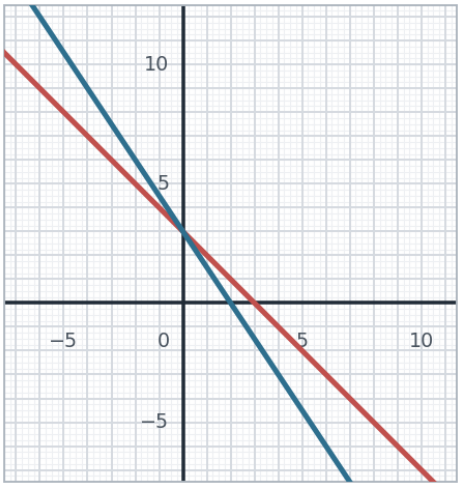
# Question 3

Which of the graphs represent the following two linear equations?

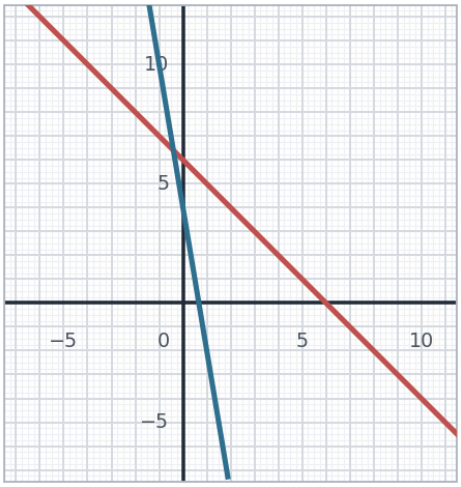
$2y + 8x = 6$

$3y + 3x = 9$

A)



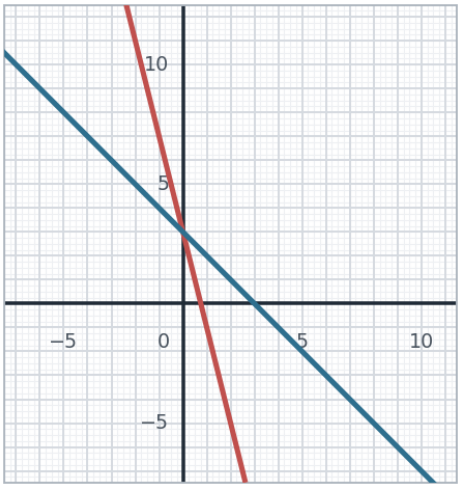

B)




C)




D)




**WORKSPACE**

# Question 4

Simplify the following expression to its simplest form.

Consider $x = 2$:

$$2x + 6y^2 + (x + 2y)(x - 3y)$$

A) 76

B) $-2y + 8$

C) 42

D) $12y^2 - 2y + 8$

**WORKSPACE**

# Question 5

Consider the following function:

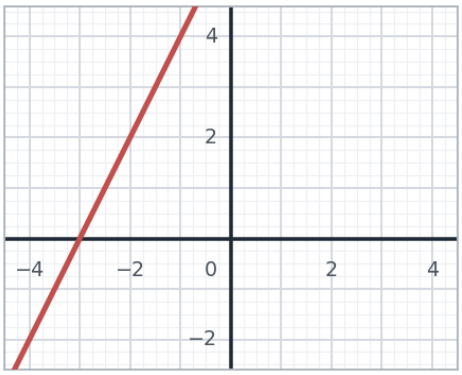
$y = 2x + 6$

**Which of the following choices represents the same relationship as the given function?**

A)

| x | y |
|---|---|
| −3 | −6 |
| 0 | 6 |
| 2 | 14 |
| 4 | 22 |

B)




C)

| x | y |
|---|---|
| 1 | 8 |
| 2 | 10 |
| 3 | 8 |
| 4 | 6 |

D) Y is the sum of 3 and twice the value of x.

**WORKSPACE**

# Question 6

What is the next number in the following series of numbers?

66 42 26 18 18 ?

A) 4

B) 8

C) 10

D) 26

**WORKSPACE**

# Question 7

What is the next number in the following series of numbers?

−1 3 2 5 7 ?

A) 6

B) 12

C) 2

D) 13

**WORKSPACE**

# Question 8

Consider the following formula:

$2E + C = 0$

**Which of the following statements is true for this formula?**

A) C always has a greater value than E.

B) C always has a lesser value than E.

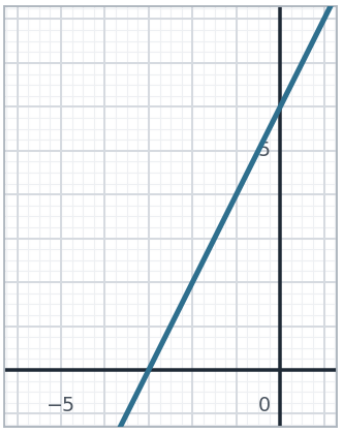
C) C always has a value equal to E.

D) C may be either greater or less than E.

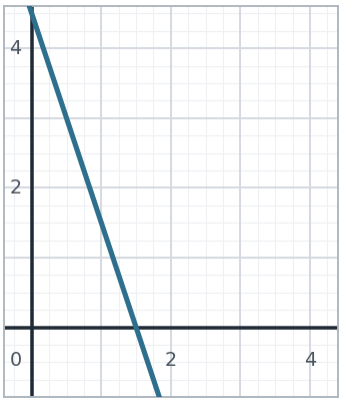
**WORKSPACE**

# Question 9

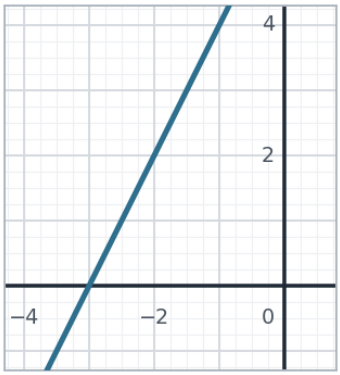
Consider the following graph of the function $y = mx + b$:




**Graph B**




**Graph D**




**Which of the following options is true in regard to this graph?**

A) Function A, $y = 5x + 6$, has the same slope and the same interception point with the y-axis.

B) Graph B is of a function with the same slope $m$ and a different interception point with the y-axis.

C) Function C, $y = 2x + 3$, has the same slope and a different interception point with the y-axis.

D) Graph D is of a function with a different slope $m$ and the same interception point with the y-axis.

# Question 10

What is the next number in the following series of numbers?

125 44 17 8 5 ?

A) −4

B) 2

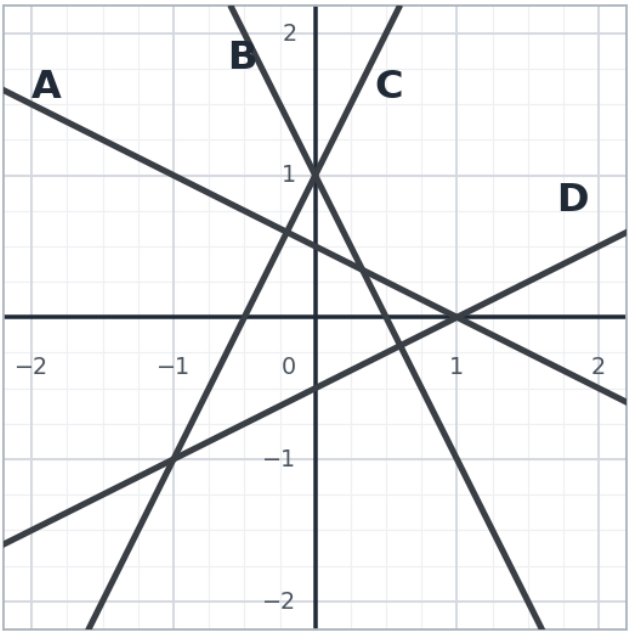
C) 4

D) 1

**WORKSPACE**

# Question 11

**Which of the following graphs represents the function $-x+\frac{y}{2}=\frac{1}{2}$?**




A) A

B) B

C) C

D) D

**WORKSPACE**

# Question 12

Consider the following formula:

$$R = \frac{T}{2} + 4$$

**Which of the following statements is true for this formula?**

A) When the value of $T$ is greater than $-10$, $R$ is positive.

B) When the value of $T$ is greater than 4, $R$ is positive.

C) When the value of $T$ is less than 4, $R$ is positive.

D) When the value of $T$ is less than $-10$, $R$ is positive.

**WORKSPACE**

# Question 13

Factor the polynomial:

$x^{14}y^{-6}z^4 - x^7y^2z^4$

A) $x^7y^{-6}z^4(x^7 - y^8)$

B) $x^7y^6z^4(x^7y^{-4})$

C) $x^7y^{-6}z^4(x^7z - y^4z)$

D) $x^{10}y^2z^3(x^4 - z)$

**WORKSPACE**

# Question 14

Consider the following formula:

$D = 2F - 12$

**Which of the following statements is true for this formula?**

A) If $F$ is greater than 2, $D$ is positive.

B) If $F$ is greater than 6, $D$ is negative.

C) If $F$ is less than 6, $D$ is negative.

D) If $F$ is less than 2, $D$ is positive.

**WORKSPACE**

# Question 15

Consider the following formula:

$$2BC^2 - 8 = A(B + A - 3)$$

**If $A = 4$ and $C = -2$, what is the value of $B$?**

A) 3

B) $-1$

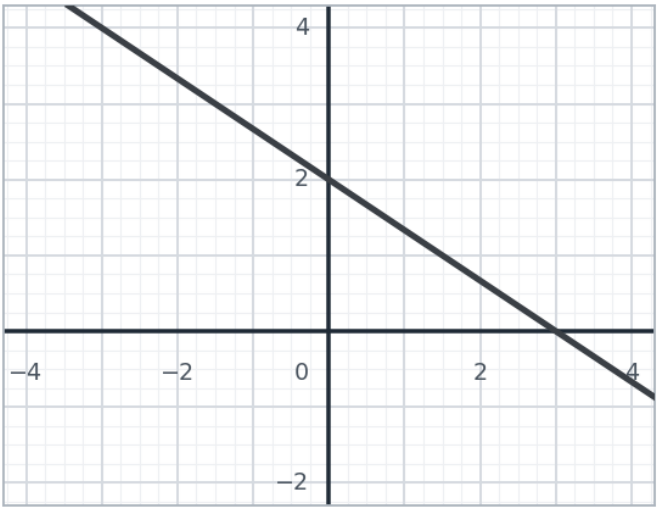
C) $-3$

D) 1

**WORKSPACE**

# Question 16

Look at the graph below. Which of the following functions does it represent?




A) $2x - 3y = 6$

B) $3x - 2y = 6$

C) $2x + 3y = 6$

D) $3x + 2y = 6$

**WORKSPACE**

# Question 17

Solve the linear equation:

$-5x - 36 = 12 + 3x$

A) 3

B) $-3$

C) $-6$

D) 8

**WORKSPACE**

# Question 18

What is the next number in the following series of numbers?

12 13 11 13 11 14 12 ?

A) 18

B) 14

C) 10

D) 16

**WORKSPACE**

# Question 19

What is the next number in the following series of numbers?

0 0 2 8 12 72 ?

A) 16

B) 78

C) 48

D) 86

**WORKSPACE**

# Question 20

What is the value of the given expression?

For $p = 5$ and $q = \frac{8p}{2}$,

$$\frac{3p+3}{2q-4} - \frac{2q-4}{3p+3} = ?$$

A) $-2.5$

B) $2.5$

C) $1.5$

D) $-1.5$

**WORKSPACE**

# Question 21

Consider the following formula:

$$\frac{A+6}{4}+\frac{C}{D}=B+14$$

**If $A=30$, $C=15$, and $D=3$, what is the value of $B$?**

A) $-4$

B) No Solutions

C) $-3$

D) 0

**WORKSPACE**

# Question 22

Simplify the following expression:

$2x(4x^5 + 2x^3 + 3x^4 - 5 + 6x^2 - x^3 - 3x^4)$

A) $8x^6 + 6x^4 + 2x^3 + 12x^2 - 5x$

B) $8x^6 + 2x^3 + 12x^2 - 10x$

C) $10x^5 + 6x^3 + 4x^2 - 10$

D) $8x^6 + 2x^4 + 12x^3 - 10x$

**WORKSPACE**

# Question 23

Simplify the expression:

$(x^{-4} - 5)(x^6 + x^4) - x^2 + x^4$

A) $5x^6 + 4x^4 + 2x^2$

B) $-5x^6 - 6x^4 + 2x^2 + 1$

C) $5x^6 - 4x^4$

D) $-5x^6 - 4x^4 + 1$

**WORKSPACE**

# Question 24

Factor the polynomial:

$x^2 - 6x + 9$

A) $(x + 3)(x - 3)$

B) $(x - 3)^2$

C) $(x + 9)(x + 1)$

D) $(x + 3)(x + 3)$

**WORKSPACE**

# Question 25

Consider the following information:

**Y is equal to twice the sum of the variable x and a constant C, where C is equal to negative four.**

**Which of the following options represents the same relationship as the aforementioned information?**

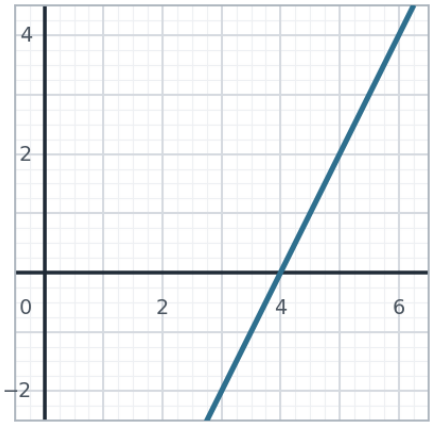
A)

| x | y |
|---|---|
| −1 | 4 |
| 0 | −8 |
| 1 | 12 |
| 2 | 16 |
| 3 | 20 |

B) X is equal to the difference between Y divided by 2 and a constant C, where C is equal to 3.

C) $y = 2x + 4$

D)




**WORKSPACE**

# Question 26

Evaluate the variable given the formulae:

$K = \frac{1}{2}mv^2$

$v = \sqrt{\frac{d}{t}}$

$K = ?$

A) $\frac{2md}{t}$

B) $\frac{md^2}{2t}$

C) $\frac{md}{2t}$

D) $\frac{md}{t}$

**WORKSPACE**

# Question 27

Consider the following formula:

$x = 2xy - 8 + y$

**If $y = 3$, which of the following statements is true for this formula?**

A) $x - y < 0$

B) $x > y$

C) $y - x < 0$

D) $x = y$

**WORKSPACE**

# Question 28

Evaluate the value of the unknown variable:

$a_m = 0.36 \qquad p = 0.6 \qquad m = 0.4$

$a_n = pm + a_m$

$a_n = ?$

A) 1.2

B) 0.6

C) 0.8

D) 2.76

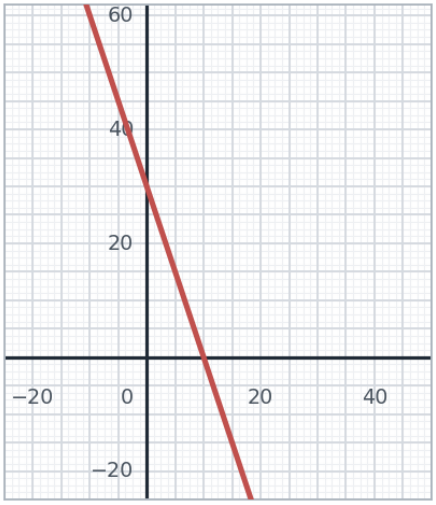
**WORKSPACE**

# Question 29

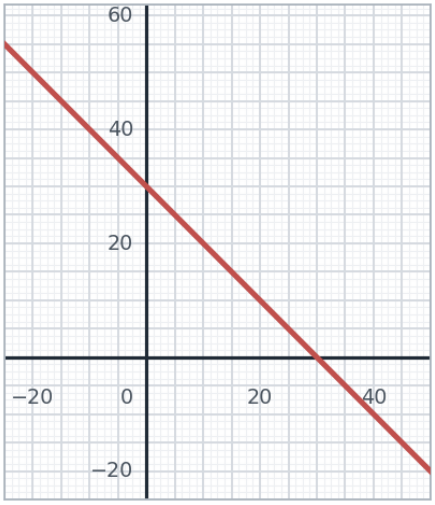
Which of the graphs represents the following table of values?

| x | y |
|---|---|
| −5 | 45 |
| 0 | 30 |
| 6 | 12 |
| 15 | −15 |

A)



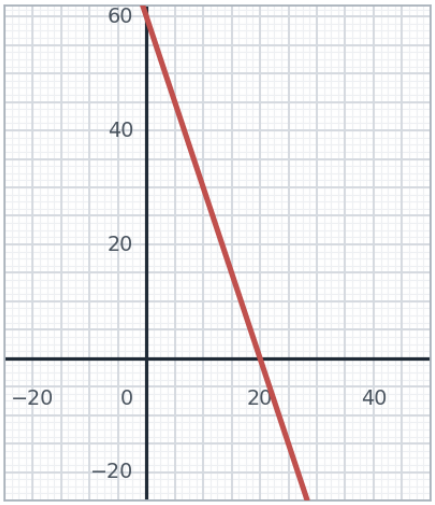

B)




C)




D)




**WORKSPACE**

# Question 30

Consider the following formula:

$$B = A \div \frac{1}{2}$$

**Which of the following statements is true for this formula (Assume B & A are not equal to zero)?**

A) $2A > B$

B) $B + \frac{1}{4} > 4A$

C) $A^2 < BA$

D) $A + 4 > B$

**WORKSPACE**

# Question 31

Isolate the variable from the formula:

$d = (g + 5)^2 - 4$

$g = ?$

A) $\sqrt{d} - 1$

B) $\sqrt{d + 4} - 5$

C) $\sqrt{d - 1}$

D) $\sqrt{d - 2} - 5$

**WORKSPACE**

# Question 32

Evaluate the value of the unknown variable:

$$\frac{2(E+G)^2}{E-J}+\frac{3(G-J)}{K}=M$$

$E=6,\ G=4,\ J=-4$, and $K=-3$. Find $M$.

A) 2

B) 12

C) 28

D) $-12$

**WORKSPACE**

# Question 33

Solve the following system of equations:

i. $13x - 26 = 7y$

ii. $17x + 9y = 34$

**Which of the following is the solution to this system?**

A) $(-1, 1)$

B) $(2, 0)$

C) $(0, 2)$

D) $(3, 1)$

**WORKSPACE**

# Answer Key

| | | | |
|---|---|---|---|
| **1. D** | **10. C** | **19. B** | **28. B** |
| **2. C** | **11. C** | **20. D** | **29. A** |
| **3. D** | **12. B** | **21. D** | **30. C** |
| **4. B** | **13. A** | **22. D** | **31. B** |
| **5. B** | **14. C** | **23. D** | **32. B** |
| **6. D** | **15. A** | **24. B** | **33. B** |
| **7. B** | **16. C** | **25. D** | |
| **8. D** | **17. C** | **26. C** | |
| **9. C** | **18. D** | **27. A** | |

**1. D** $2(4+A)+3=5(-7) \rightarrow 8+2A+3=-35 \rightarrow 2A=-46 \rightarrow A=-23$

**2. C** From $x=1$ to $x=3$, $y$ goes up 6, so it rises 3 per 1: $y=3x+8$.
Table C: $3(-5)+8=-7$, $3(-3)+8=-1$, $3(0)+8=8$, $3(5)+8=23$ ✓

**3. D** Get $y$ alone: $y=3-4x$ and $y=3-x$. Both cross the y-axis at 3.
Red falls 4 per 1 (hits the x-axis at $\frac{3}{4}$); blue falls 1 per 1 (hits it at 3). That's D.

**4. B** $4+6y^2+(2+2y)(2-3y)=4+6y^2+4-6y+4y-6y^2=-2y+8$

**5. B** The graph crosses the x-axis at $-3$ and rises 2 per 1: $y=2x+6$ ✓
Table A is $y=4x+6$. D says $y=2x+3$.

**6. D** Differences: $-24$, $-16$, $-8$, 0 (each 8 more). Next is $+8$: $18+8=26$

**7. B** Each number is the sum of the two before it: $-1+3=2$, $3+2=5$, $2+5=7$, $5+7=12$

**8. D** $C=-2E$. If $E=1$, $C=-2$ (less). If $E=-1$, $C=2$ (greater).

**9. C** The line crosses the x-axis at $-3$ and rises 2 per 1: $y=2x+6$.
$y=2x+3$ has the same slope (2) and a different y-intercept (3) ✓
Graph B falls (different slope). Graph D is the same line ($2x+6$), so 'different slope' is false.

**10. C** Differences: $-81$, $-27$, $-9$, $-3$ (each divided by 3). Next is $-1$: $5-1=4$

**11. C** Multiply by 2: $-2x+y=1 \rightarrow y=2x+1$.
Line C crosses the y-axis at 1 and rises 2 per 1.

# Answer Key (cont.)

**12. B** $R = 0$ at $T = -8$; above $-8$, $R$ is positive — so above 4 it's positive ✓
A fails at $T = -9$ ($R = -\frac{1}{2}$).

**13. A** Pull out the smallest powers: $x^7$, $y^{-6}$, $z^4$.
$x^{14} \div x^7 = x^7$ and $y^2 \div y^{-6} = y^8$ → $x^7 y^{-6} z^4 (x^7 - y^8)$

**14. C** $D = 0$ at $F = 6$. Below 6, $D$ is negative ✓. A fails at $F = 3$ ($D = -6$).

**15. A** $8B - 8 = 4(B + 1)$ → $8B - 8 = 4B + 4$ → $4B = 12$ → $B = 3$

**16. C** The line crosses at (0, 2) and (3, 0). Check: $2(0) + 3(2) = 6$ ✓ and $2(3) + 3(0) = 6$ ✓

**17. C** $-8x = 48$ → $x = -6$

**18. D** Steps: +1, −2, +2, −2, +3, −2. The 'up' steps go 1, 2, 3, so the next is +4: $12 + 4 = 16$
(This one is tricky — the pattern only shows up when you split the up-steps from the −2s.)

**19. B** Alternates multiply then add with the same number: ×2, +2, ×4, +4, ×6, +6
$0 \times 2 = 0$, $0 + 2 = 2$, $2 \times 4 = 8$, $8 + 4 = 12$, $12 \times 6 = 72$, $72 + 6 = 78$

**20. D** $q = 20$. $\frac{18}{36} - \frac{36}{18} = \frac{1}{2} - 2 = -1.5$

**21. D** $\frac{36}{4} + \frac{15}{3} = 9 + 5 = 14 = B + 14$ → $B = 0$

**22. D** Inside first: $4x^5 + x^3 + 6x^2 - 5$ (the $3x^4$'s cancel).
Times $2x$: $8x^6 + 2x^4 + 12x^3 - 10x$

**23. D** FOIL: $x^2 + 1 - 5x^6 - 5x^4$. Then $-x^2 + x^4$:
$-5x^6 - 4x^4 + 1$

**24. B** Two numbers that multiply to 9 and add to −6: −3 and −3. So $(x - 3)^2$.

**25. D** $y = 2(x + (-4)) = 2x - 8$: crosses the x-axis at 4 and rises 2 per 1. That's the graph.
B works out to $y = 2x + 6$; C is $2x + 4$.

**26. C** $v^2 = \frac{d}{t}$ (squaring undoes the root). $K = \frac{1}{2} \cdot m \cdot \frac{d}{t} = \frac{md}{2t}$

# Answer Key (cont.)

**27. A** $x = 6x - 8 + 3 \rightarrow -5x = -5 \rightarrow x = 1$. $x - y = 1 - 3 = -2 < 0$ ✓

**28. B** $0.6 \times 0.4 = 0.24$. $0.24 + 0.36 = 0.6$

**29. A** Slope: $\frac{12 - 30}{6 - 0} = -3$. Crosses the y-axis at 30: $y = 30 - 3x$, x-intercept 10.
Graph A crosses at (0, 30) and (10, 0). B is too shallow; C crosses at 15; D at 60.

**30. C** Dividing by $\frac{1}{2}$ doubles: $B = 2A$. $BA = 2A^2$, which is always bigger than $A^2$ (since $A \neq 0$) ✓
A: $2A = B$, not greater. D fails at $A = 5$ ($9 > 10$ is false).

**31. B** Add 4: $d + 4 = (g + 5)^2$. Square root: $\sqrt{d + 4} = g + 5$. Subtract 5.

**32. B** $\frac{2(10)^2}{10} + \frac{3(8)}{-3} = 20 - 8 = 12$

**33. B** Fastest: test the choices. (2, 0): $26 - 26 = 0$ ✓ and $34 + 0 = 34$ ✓